

Progress meeting

Kernel regression experiments

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Experiment setup

- Inputs lie on the circle:

$$x(\gamma) = (\cos \gamma, \sin \gamma), \quad \gamma \in [0, 2\pi).$$

- We replace the network by its *infinite-width kernel limit*.
- Regression kernel = full NTK Gram matrix

$$\Theta_{XX} = [\Theta(\gamma_i, \gamma_j)]_{i,j=1}^n.$$

- Initialization of predictions: $y_{\text{pred},0} = 0$
- Training is kernel gradient descent for least-squares:

$$y_{\text{pred},t+1} = y_{\text{pred},t} - \eta \Theta_{XX}(y_{\text{pred},t} - y).$$

What we want to validate

Residual dynamics are linear:

$$r_t = y_{\text{pred},t} - y, \quad r_{t+1} = (I - \eta \Theta_{XX}) r_t.$$

If

$$\Theta_{XX} = Q \Lambda Q^\top,$$

then each empirical eigenmode evolves independently:

$$\alpha_k(t+1) = (1 - \eta \lambda_k) \alpha_k(t), \quad \alpha_k(t) = \langle r_t, q_k \rangle.$$

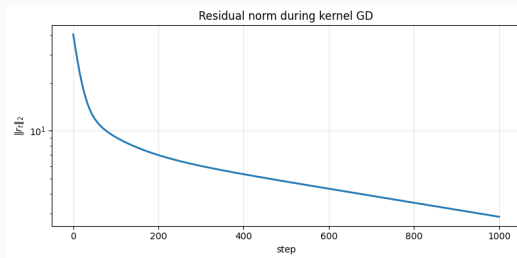
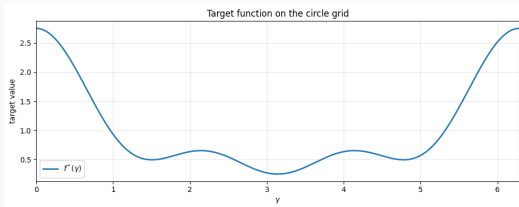
For small η , this is approximately exponential:

$$\alpha_k(t) \approx e^{-\eta \lambda_k t} \alpha_k(0).$$

Main checks:

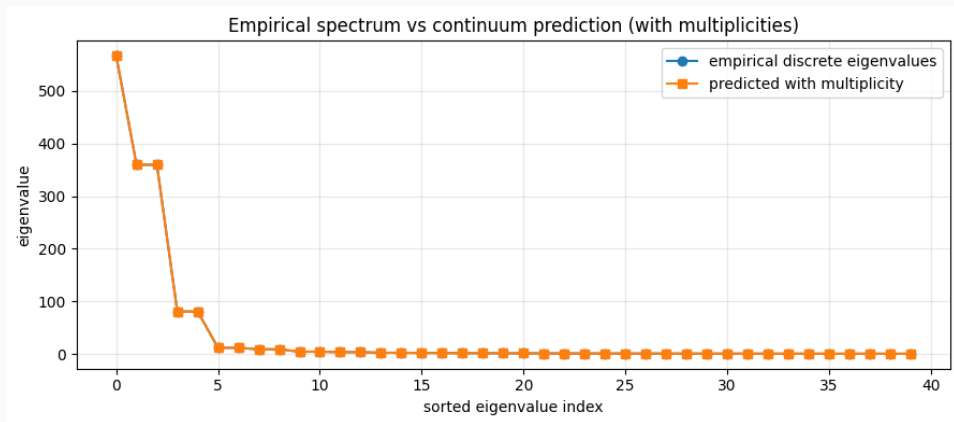
- Does the empirical discrete spectrum match the continuum prediction?
- Do observed mode-wise decay rates match the ones predicted by the eigenvalues?
- Do lower-frequency modes decay faster than higher-frequency ones?

Target and residual decay



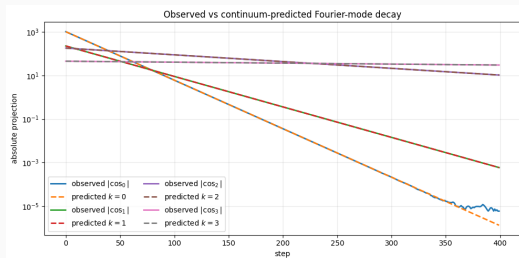
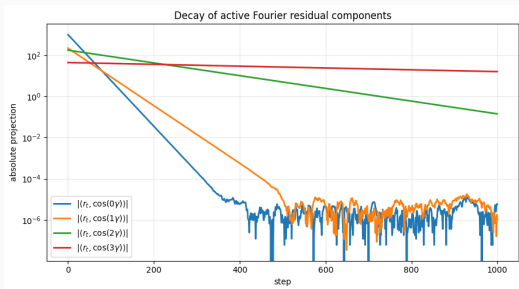
- Left: target function on the circle grid. $f^*(\gamma) = 1 + 0.9 \cos(\gamma) + 0.5 \cos(2\gamma) + 0.35 \cos(3\gamma)$.
- Right: residual norm decays smoothly under kernel GD.

Empirical spectrum vs continuum prediction



The empirical NTK spectrum matches the continuum prediction well once the multiplicity of Fourier modes is accounted for.

Mode-wise decay



- Left: observed decay of the active residual components.
- Right: observed decay compared with the ideal rate predicted by the spectrum.

Current takeaways

- The analytic full-NTK kernel on S^1 is implemented and numerically consistent.
- Kernel regression with NTK dynamics produces the expected residual decay.
- The empirical spectrum matches the continuum Fourier prediction on an evenly spaced grid.
- Mode-wise decay follows the spectral prediction: lower-frequency / larger-eigenvalue components decay faster.

Next steps

- Initialization of predictions given by the GP / NNGP limit:

$$y_{\text{pred},0} \sim \mathcal{N}(0, K_{0,XX}).$$

- Move from theory-validation runs to sampled finite- n experiments.